Nguyễn Văn Tiệm

nguyentiem21101998@gmail.com - 0355180555

PROFESSIONAL SUMMARY

- >5 years of experience in embedded system software development.
- >4 years of experience in IoT design and development with big US and European companies.
- Highly confident in teamwork, risk management.

TECHNICAL EXPERTISE

Program languages	C, C++, Java, Python.
Coding standards	MISRA C
Microcontrollers	AVR, ESP32, Silabs, Nordic, STM32, Toshiba, Abov, Renesas …
Peripherals	UART, I2C, SPI, ADC, DAC, DMA, Timer, PWM, Interrupt (NVIC)
IoT technologies and protocols	HTTP, ZIGBEE, Z-WAVE, BLE, OpenThread, MATTER, SIDEWALK, Websocket, MQTT.
RTOS	FreeRTOS, ZephyrOS
Source Control Tools	GIT (github, gitlab platform), SVN
Task/Bug Management Tools	Jira
IDE/Utilities	Keil, GCC ARM, Simplicity Studio 5, Visual Code, nRF Connect, Stm32Cube IDE, ESP IDF, Android Studio, IAR, E2 Studio, Arduino IDE.
Operating Systems	Windows, Linux
Hardware design	Proteus
Data Structure and Algorithms	BFS, DFS, backtracking, Heap, Stack, Queue, Tree, Tries, Priority Queue, Map, Hash.
Certificate	STP level pro(Samsung talent programmer), SPCT (LGEDV)
Languages	Read and understand documents, communicate in English at a basic level.

PROFESSIONAL EXPERIENCE

04/2025 – present

Company: LGEDV

Project Description: Develop new feature for some washing machine models include wash-only , wash towel, wash and dry (Heat pump).

Position: Main Embedded developer , reviewer

Responsibility:

- Development: Implement code base on the product requirements document.
  - Analyze legacy code flows and design new logic that does not impact existing workflows, ensuring backward compatibility with previous versions.
  - Store user configuration data in EEPROM.
  - Synchronize machine status and selected user information to the server.
  - Write test cases and execute testing to ensure quality before handover to the SQE team.
  - Write documents.
  - Fix remaining bugs in legacy versions.
  - Review code for team members.
- Managements:
  - Gitlab, SVN, Collab, Jira

Techniques:

- C language, python
- Microcontroller : Above, Toshiba, Renesas.
- IAR, E2 Studio.

12/2023 – 04/2025	**Company:** FPT Software – Hanoi, Vietnam. **Project Description:** Develop a garage door integrated with BLE, Wi-Fi, and Matter wireless technologies. **Position**: Embedded Developer **Responsibility**: • Development: - Analyze the hardware components of the device from the schematic. - Define device tree for custom board of project. - Bring up the board base on sample example. - Implement OTA via Bluetooth with custom external flash. - Implement control of LED behavior. - Launch Matter Hub on Raspberry Pi 4. - Test OTA via Matter over Thread - Implement the RTC module using RX8130CE (I2C) - Implement security algorithms (ECDH) to establish a secure channel for BLE - Design the BLE message format. - Implement transfer message via BLE. - Implement a FAT and LittleFS file system to manager users, door configurations, and logs. - Port matter profiles, switching between the Door lock profile and window covering profile. - Implement UART communication. - Implement MQTT with AWS IoT Core. - Optimize the source code to reduce RAM and Flash usage. - Implement OTA for mother board use BLE and UART - Enable Thread shell for testing - Implement safety reboot, auto reboot for system - Optimize mutex, queue, and buffer. • Management: - Jira and Git **Techniques**: - C, C++, Python, Bash shell, Windows Batch, Makefile - Docker, Linux, Cygwin, Raspberry Pi, Atlassian Jira/BitBucket/Confluence, nRF Connect, Ozone debugger. - BLE, Wi-Fi, Matter, ZephyrOS, UART, I2C, SPI
10/2023 – 11/2023	**Company:** FPT Software – Hanoi, Vietnam. **Project Description:** Embedded IoT Engineer - Maintain Project, Study and Seminar **Position**: Embedded Engineer **Responsibility**: • Study: - Nordic SDK and related platforms. - Learn about the Zigbee technology, implement switch on/of and light cluster on Dev board. - Learn about Side Walk technology. - Learn about Matter over Thread technology, and implement Door lock profile on Silicon Labs board. - Launch Matter Hub on Raspberry Pi 4 and Laptop. - Build bootloader, launch Linux OS on beagle bone.

	- Fix issues, and upgrade SDK for door lock project. • Management: - Jira and Git **Techniques**: - C, C++. - Docker, Linux, Cygwin, Raspberry Pi, Github, Simplicity Studio, SLCC, SLCP,… - Zigbee, Side Walk, Matter over Thread, Matter Over Wi-Fi, IoT network, RTOS
04/2023 – 09/2023	**Company:** FPT Software – Hanoi, Vietnam. **Project Description:** Develop a Door Lock integrated with Z-Wave wireless technology. **Position**: Embedded Engineer **Responsibility**: • Development: - Analyze and optimize power consumption. - Implement UART communication. - Compile commands from customer and document them. - Map Z-Wave command class to TLV command and vice versa. - Read and analyze source from customer for integrate test. - Set up Z-sniffer for testing purpose. - Set up self-test tools to get Z-Wave certificate. • Management: **Techniques**: - C, Python. - RTOS - Git, Simplicity Studio, Ozone debugger, Logic Analyzer , SLCC, SLCP,…
04/2021 – 03/2023	**Company:** Samsung Vietnam Mobile Centre – Hanoi, Vietnam. **Project Description:** Software Engineer **Position**: Android Developer **Responsibility**: • Development: - Maintain Samsung's Application - Develop a PDF reader application - Develop an automated testing application **Techniques**: - Java, Kotlin. - Android Activity, android fragment, threads, services, broadcast receiver, content provider, SQL lite, Rx java, design pattern.
Other	• Develop custom bootloader, FOTA for STM32. • Develop a keypad driver for STM32U585 using IQS7222B for touch scanning and TCA8418 for button LEDs via I2C. • Develop multiple drivers that use I2C communication, including MPU6050, LIS2, HTS221, and LPS22. • Implement peripheral communication using DMA on STM32F411. • Implement an audio project using I2S with ESP32 and the ESP-IDF SDK. • Develop a driver for the LCD G9A01 using SPI. • Implement a driver for external NOR flash Winbond 25Q64 via SPI.

	• Implement OTA via BLE for nRF52840 using both internal and external flash. • Design and develop a smart home project using WebSocket, including firmware development for ESP8266, WebSocket server development with Node.js, and an Android app development in Java.
EDUCATION	
2016 - 2021	**Computer Engineering** *Hanoi University of Science and Technology, Vietnam*